



11534 - Non-equilibrium processes in young stellar objects: comprehensive modeling of mid-infrared spectra

Cycle: 5, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Phillip C. Stancil (PI)	University of Georgia Research Foundation, Inc.
Ziwei Zhang (CoI) (CoPI)	RIKEN Wako Institute
Mr. James Squires (CoI) (CoPI)	University of Georgia Research Foundation, Inc.
Ms. Anna Sophia Mehta (CoI) (CoPI)	University of Georgia Research Foundation, Inc.
Ms. Delaney Murray (CoI) (CoPI) (Contact)	University of Georgia Research Foundation, Inc.

OBSERVATIONS

ABSTRACT

Rovibrational molecular emission from protostellar systems has been extensively observed with JWST/MIRI. A range of molecules have been detected, many providing unprecedented diagnostics of individual physical components of the protostellar source and outflow. However, most analyses have focused on spectroscopic fits to equilibrium models and parametric dust continuum fitting. Such approaches give a first look at protostellar properties and local environmental parameters, but may compromise scientific insight as a consequence of implicit approximations. We propose a multistep, multiprocess approach that includes self-consistent modeling of non-equilibrium gas-phase chemistry, dust opacity and extinction, thermal balance including all relevant heating/cooling processes, non-equilibrium rovibrational molecular level populations, and 1D radiative transfer. The latter two components constitute a non-local thermodynamic equilibrium (LTE) approach which we have built from advanced rovibrationally-resolved molecular excitation models for H₂, OH, CO, SiO, SiS, SO, CS, H₂O, SiS, and SO₂. We will first perform fits of MIRI observed spectra to non-LTE models using RADEX to estimate environment parameters improving upon prior LTE models and guiding more comprehensive modeling. We will apply Cloudy for a comprehensive self-consistent modeling of all molecules and the dust continuum. We will use archival data for two low-mass (IRAS2A and L1527) and two high-mass (IRAS 23385+6053 and IRAS 20126+4104) sources. Finally, the modeling

will extend to other wavelength ranges taken from JWST/NIRSpec, Spitzer, Herschel, and ALMA for a complete inventory of molecular features.

OBSERVING DESCRIPTION